Dataset Familiarization Summary: (**Entire Folder: Week 1 Data Exploration & EDA Report)**

The dataset represents DePaul University's international admissions tracking system, capturing the full pipeline from application submission to outreach and follow-up activity. Each row does not strictly represent a unique applicant — 2,543 Reference IDs appear more than once, indicating that the dataset is structured around outreach events or campaign interactions (one applicant may have multiple rows for multiple contact attempts).

Structure Type: This is a wide, denormalized dataset that appears to have been assembled by joining at least three source tables: (1) a core applicant profile table, (2) an outreach/contact activity log, and (3) a campaign/template management table. Evidence of this is the repeated columns with suffixes (-2, -3) such as Reference_ID-2, Recieved_At-2, and Created_At-3.

Variable Types: The 80 columns span: free-text identifiers (names, addresses), categorical variables (Country, College, Outcome), timestamp fields (dates), binary flags (Slate_Form_Filled, Escalation_Required), and numeric identifiers (IDs, Campaign IDs). Notably, 27 columns are entirely empty (100% missing), indicating schema fields that were included in the export but not populated for this cohort.

Primary Focus: The dataset primarily covers applicants from India (61.2%), the United States (10.1%), Ghana (5.8%), and Nigeria (5.7%). All records are for DePaul University, and the majority are prospective or admitted graduate students.

Data Dictionary (Spreadsheet Format):

The table below documents all 80 fields. The Group column categorizes fields into logical clusters: Applicant Profile, Academic, Address, Outreach/Activity, Campaign, and System. Fields marked as empty (100% null) are flagged accordingly.

DePaul_Data_Dictionary.xlsx

Data Quality Report:

27 Columns are 100% Empty. Fields such as Major, Degree_Type, Citizenship, Status, Date_of_Birth, Is_Admitted, SEVIS_ID, and others are completely unpopulated in this extract. These should be dropped before any analysis or modeling to reduce dimensionality noise.

Non-Unique Primary Key. Reference_ID is not unique — 2,543 IDs repeat, one row per outreach event. Analysts must aggregate to unique applicant level before computing applicant-level statistics.

Structural Duplicates from Join. Columns suffixed -2 and -3 (Reference_ID-2, University-2, University-3, Created_At-2, etc.) are artifacts of table joins and are redundant or near-constant. They should be removed or renamed.

Column Naming Issues. "Recieved_At" is misspelled (should be "Received_At"). "Counsler" should be "Counselor". Standardize all column names to snake_case for downstream processing.

Phone Number Format Inconsistency. Phone numbers appear in multiple formats: international prefixes ("91 8125376437"), plain numbers, and scientific notation ("9.20E+11"). Requires regex normalization before use.

Country Name Casing. Country values are inconsistently capitalized (e.g., "United states", "United kingdom" vs mixed-case). Apply .str.title() or a mapping table to standardize.

Batch Timestamp Fields. Created_At, Modified_At, Created_At-2, Modified_At-2, Created_At-3 have only 1–2 unique values each, indicating they captured the batch export time rather than individual record timestamps. These should not be used for time-series analysis.

Intake Field Requires Parsing. The Intake column embeds three variables (Program, Degree, Term) in a single string. Extract program name, degree type, year, and season into separate columns for analysis.

Exploratory Data Analysis (EDA) & Visualizations:

Country Distribution

Top countries represented:

Country	Approx. Count
India	4,617
United States	764
Ghana	440
Nigeria	430
Pakistan	273

Insight

India accounts for more than half of all records, indicating the institution's strong recruitment presence there.

University Distribution

University	Count

DePaul University	7,543

Insight

All records belong to the same institution.

Top First-Choice Colleges

College	Count
Jarvis College of Computing and Digital Media	3,831
Kellstadt Graduate School of Business	2,523
College of Science and Health	269
College of Education	208
College of Liberal Arts and Social Sciences	207

Insight

Computing and business programs dominate student interest.

Most Popular Intakes

Examples:

Intake	Count
Computer Science – MS (Fall 2024)	504
Business Analytics – MS (Fall 2024)	419
Data Science – MS (Fall 2024)	254
Cybersecurity – MS (Fall 2024)	181
Artificial Intelligence – MS (Fall 2024)	160

Insight

Technology-focused graduate programs account for a large share of applications.

Application Source

The dataset contains only two application source categories.

Insight

Further analysis should compare admission outcomes between sources.

Insights Summary:

- India is the dominant feeder country, accounting for 61.2% of all applicants. The next three countries — the US (10.1%), Ghana (5.8%), and Nigeria (5.7%) — collectively represent less than 22%. Any segmented outreach strategy must treat India as a standalone priority segment with potentially distinct communication strategies.
- STEM and Business programs drive the vast majority of applications. Business Analytics, Computer Science, and Data Science alone account for 2,675 applicants (~35% of all records). The Jarvis College of Computing and Kellstadt Graduate School together absorb 85%+ of all applicants. This concentration has implications for counselor workload and resource allocation.
- Fall intake is overwhelmingly dominant. Fall 2024 alone accounts for 3,284 applicants (43.5% of all records). Fall intakes across all years account for over 65% of total volume. Winter and Spring intakes play a supporting role (~25%). Planning and staffing should peak in the summer-to-fall window.
- The dataset captures outreach events, not unique applicants. 2,543 of 5,000 unique Reference IDs appear more than once. The 7,543 rows represent multiple contact events per applicant. Counting rows as applicants would significantly overstate the applicant pool. All applicant-level KPIs must be computed after deduplication by Reference_ID.
- The most common outreach result is "Follow up" — suggesting most first contacts don't resolve the issue. 1,375 outreach events ended in "Follow up" and 1,177 in "Missing 0 Transcript (NOT Study Group)", indicating transcript submission is a major bottleneck in the admissions pipeline. This is a strong candidate for a targeted workflow improvement.
- Outreach is concentrated among just 4 counselors, with one handling 48% of contacts. Jagdeep alone handled 2,085 of 4,349 total contact events. This creates significant key-person risk. A balanced distribution analysis across counselors could inform staffing decisions.
- 27 of 80 columns (33.75%) are completely empty and should be dropped. Including fields like Date_of_Birth, Is_Admitted, Citizenship, SEVIS_ID, and all secondary outreach fields. These inflate the schema and add no analytical value for this cohort. After removing empty and redundant columns, a lean working dataset of ~25–30 meaningful fields can be assembled.
- The dataset is a denormalized join of three logical tables. A cleaner approach for Week 2 would be to decompose it into: (1) Applicant Master Table (5,000 unique records), (2) Outreach Activity Log (4,349 events), and (3) Campaign Reference Table (10 campaigns). This normalized structure will enable more accurate aggregation and cross-table analysis.
- Admission volume peaked in 2024 (3,805 admitted). The 2025 cohort shows only 607 admits so far — but this likely reflects the in-progress nature of 2025 applications rather than a decline. Tracking admit rates over time (once Is_Admitted is populated) would provide conversion funnel insights from application to admit.

Visual Representation of DePaul Dataset